

CARLOS LORENZO-ZÚÑIGA MARÍ

Valencia, ES · contact@carloslorenzo.dev · +34 722599608 · [LinkedIn](#) · carloslorenzo.dev

EDUCATION

- **Universitat Politècnica de València** Valencia, ES
Degree in Biomedical Engineering *Sep 2025 – Jun 2029*
 - **Focus Areas:** Embedded Systems, Biosignal Processing, Neurotechnology.

EXPERIENCE

- **Daidalonic** Valencia, ES
Signal Processing Engineer *Oct 2025 – Present*
 - **Hardware Engineering:** Engineered 2 multi-layer sEMG data acquisition PCBs incorporating galvanic isolation and ESD protection, enabling ~99% muscle contraction detection accuracy in downstream ML models (Random Forest/CNNs).
 - **Analog Front-End Design:** Designed 2-stage active analog front-ends featuring 30-300Hz bandpass filtering and DRL bias, achieving up to 1100V/V amplification and robust RMS/offset signal conditioning.
 - **Hardware Validation:** Standardized hardware validation by authoring DVT protocols, expanding testing bandwidth across the team and successfully isolating and resolving a critical power supply reliability issue.
- **Caxton College** Puzol, ES
IT Specialist *Jun 2025 – Jul 2025*
 - **Infrastructure Automation:** Automated a domain controller (Active Directory) migration for 1,700 students, reducing a 3-day multi-technician process to 30 minutes and saving thousands of euros in labor costs.
 - **Custom Tooling:** Developed a custom Google Apps Script platform for faculty to report and monitor over 1,000 students, streamlining communication workflows via database fuzzy-finding and automated emailing.

PROJECTS

- **8-Channel EEG Headset (Bespoke BCI):** Developed a bespoke 8-channel EEG headset utilizing a Texas Instruments ADS1299 and an ESP32-S3 on a custom 4-layer PCB, reducing prototype costs to under 250 EUR (<10% of comparable industry models). Implemented active bias (DRL) and digital filtering (50Hz notch, 4-38Hz bandpass) alongside a bespoke ESP-IDF driver. Deployed a quantized EEGNet ML model running at 10Hz, achieving 70% motor imagery accuracy on public datasets.
- **High-Performance C++ Physics Engine:** Programmed a custom physics engine from scratch in C++ capable of rendering complex multi-body rigid collisions (20+ objects in high-density environments), maintaining a seamless 144Hz refresh rate.

TECHNICAL SKILLS

- **Hardware & Embedded:** PCB Design (KiCad, EasyEDA), ESP-IDF Firmware, Microcontrollers (ESP32-S3), ADS1299, Soldering, Oscilloscopes, Function Generators, DVT protocols.
- **Signal Processing:** sEMG/EEG Acquisition, Analog/Digital Filtering, Amplification, DRL, Galvanic Isolation.
- **Software & Languages:** C/C++, Python (PyTorch, NumPy, Pandas), C#, TypeScript, Git, Linux.
- **Languages:** Spanish (Native), English (C2 Cambridge), German (B2 Goethe).